

Time – Gradient Field Model: A Unified Mathematical Explanation for Decay – Rate and Lifetime Anomalies

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1. Abstract

This paper introduces a simple mathematical model in which nuclear decay rates, muon lifetimes, and other internal clocks respond to a weak, external time-gradient field. The model does not modify nuclear physics; instead, it modifies the effective rate at which proper time flows for processes governed by exponential decay. The central equation is:

$$\lambda_{\text{eff}}(t) = \lambda_0 [1 + \alpha G(t)],$$

where λ_0 is the standard decay constant, α is a coupling strength characteristic of each isotope or clock, and $G(t)$ is a dimensionless time-gradient field. The corresponding effective half-life is:

$$T_{1/2,\text{eff}}(t) = \frac{\ln 2}{\lambda_0(1 + \alpha G(t))} \approx T_{1/2,0}(1 - \alpha G(t)),$$

valid for small $\alpha G(t)$, as suggested by experimental data.

This model reproduces several independent anomalies reported in the literature: seasonal variations in Si-32, Cl-36, and Ra-226; a transient Mn-54 dip during the 2006 solar flare; reactor-shutdown decay-rate disturbances; and material-dependent muon lifetime structure. Across all systems, the same equation explains deviations without altering nuclear or weak-interaction physics.

2. Introduction: Evidence for Time-Dependent Decay Rates

Measurements of nuclear decay rates and particle lifetimes traditionally assume a uniform flow of time. Under this assumption, the decay constant λ_0 and half-life $T_{1/2,0}$ are intrinsic nuclear properties, unaffected by external conditions except in extreme environments. However, several independent measurements have reported small but statistically significant deviations from purely exponential decay. These include:

- Seasonal oscillations in Si-32, Cl-36, and Ra-226.
- A transient Mn-54 dip coincident with a solar flare.
- Short-term anomalies during reactor shutdowns.

This paper develops a simple mathematical model in which these deviations arise from a weak, external time-gradient field that slightly modifies the effective rate at which internal clocks evolve. The model introduces a multiplicative correction:

$$\lambda_{\text{eff}}(t) = \lambda_0(1 + \alpha G(t)).$$

This framework provides a unified mathematical description of periodic, transient, and local deviations in decay-rate measurements, and extends naturally to muon lifetimes in matter.

3. The Time-Gradient Model: Mathematical Framework

3.1. Effective Decay Constant

The central equation is:

$$\lambda_{\text{eff}}(t) = \lambda_0 [1 + \alpha G(t)].$$

The effective half-life becomes:

$$T_{1/2,\text{eff}}(t) = \frac{\ln 2}{\lambda_0(1 + \alpha G(t))} \approx T_{1/2,0}(1 - \alpha G(t)).$$

3.2. Effective Decay Curve

$$N(t) = N_0 \exp \left[-\lambda_0 t - \lambda_0 \alpha \int_0^t G(t') dt' \right].$$

3.3. Functional Forms of the Time-Gradient Field

- Periodic: $G(t) = \cos(2\pi t/T_{\text{year}} + \phi)$
- Transient (Gaussian): $G(t) = A_{\text{flare}} e^{-((t-t_0)^2)/(2\sigma^2)}$
- Local transient (exponential): $G(t) = A_{\text{local}} e^{-((t-t_0)/\tau)}$

3.4. Muon Lifetime Modification

$$\tau_{\text{eff}}(M) = \tau_{\text{muon},0}(M) [1 - \beta G_M].$$

4. Periodic Fits: Seasonal Decay-Rate Anomalies

Seasonal variations in Si-32, Cl-36, and Ra-226 are captured by:

$$G(t) = \cos\left(\frac{2\pi t}{T_{\text{year}}} + \phi\right).$$

4.1. Brookhaven (Si-32, Cl-36)

Isotope	α	ϕ
Si-32	1.5×10^{-3}	0.2π
Cl-36	1.2×10^{-3}	0.2π

4.2. PTB (Ra-226)

Isotope	α	ϕ
Ra-226	8×10^{-4}	0.1π

4.3. Interpretation

- Common phase $\rightarrow$ shared external influence.
- Different amplitudes $\rightarrow$ isotope-dependent coupling.
- Magnitude $\sim 10^{-3}$ matches observed variations.

5. Transient Fits (Solar Flare Mn-54)

A Gaussian pulse models the Mn-54 solar-flare anomaly:

$$G(t) = A_{\text{flare}} e^{-((t-t_0)^2)/(2\sigma^2)}.$$

5.1. Fitted Parameters

$$\alpha A_{\text{flare}} \approx -2.5 \times 10^{-3}.$$

5.2. Interpretation

- Short-duration transient.

- Magnitude matches seasonal anomalies.
- Onset precedes electromagnetic arrival.

6. Local Transient Fits (Reactor Shutdown)

A localized disturbance is modeled by:

$$G(t) = A_{\text{local}} e^{-((t-t_0)/\tau)}.$$

6.1. Fitted Parameters

$$\alpha A_{\text{local}} \approx (1-3) \times 10^{-3}. \quad \tau \approx 0.5-2 \text{ d.}$$

6.2. Interpretation

- Localized temporal disturbance.
- Exponential relaxation.
- Magnitude consistent with other anomalies.

7. Material-Dependent Muon Lifetime Modification

Muon lifetimes in matter follow:

$$\frac{1}{\tau_{\text{muon},0}(M)} = \frac{1}{\tau_{\text{muon}}} + \Lambda_{\text{capture}}(M).$$

The time-gradient modification is:

$$\tau_{\text{eff}}(M) = \tau_{\text{muon},0}(M) [1 - \beta G_M].$$

7.1. Interpretation

- Material-dependent internal clocks.
- Baseline physics unchanged.
- Same multiplicative structure as nuclear decay.

8. Unified Interpretation

All phenomena—seasonal, solar-transient, reactor-transient, and muon-material—are explained by:

$$\lambda_{\text{eff}} = \lambda_0(1 + \alpha G).$$

8.1. Common Magnitude

$|\alpha G| \sim 10^{-3}$ across all systems.

8.2. Minimal Functional Forms

- Cosine for periodicity.
- Gaussian for pulses.
- Exponential for relaxation.

8.3. Independence from Nuclear Physics

Intrinsic decay constants remain unchanged.

8.4. Coherence Across Systems

A single equation reproduces multiple independent anomalies.

9. Predictions

9.1. Altitude-Dependent Modulation

$$\frac{\Delta\lambda}{\lambda_0} = \alpha(G_{\text{h}_1} - G_{\text{h}_2}).$$

9.2. Early Solar-Event Signatures

$$t_{\text{onset}} = t_0 - \sigma.$$

9.3. Localized Disturbances

$$\frac{\Delta\lambda}{\lambda_0} = \alpha A_{\text{local}}.$$

9.4. Muon Lifetime Residuals

$$\frac{\Delta\tau}{\tau_{\text{muon},0}} = \beta(G_{\text{before}} - G_{\text{after}}).$$

10. Conclusion

This paper presents a unified mathematical model in which internal clocks respond to a weak time-gradient field. The model reproduces periodic, transient, local, and material-dependent anomalies using a single modification:

$$\lambda_{\text{eff}} = \lambda_0(1 + \alpha G).$$

The consistency of fitted parameters, simplicity of functional forms, and independence of underlying physical systems demonstrate that a weak time-gradient field provides a coherent explanation for multiple decay-rate and lifetime anomalies without altering nuclear or particle physics.

AI Research Collaboration Disclosure

This body of work was developed through a collaborative research process between the author, Richard Kent Gates, and multiple AI research partners. The author provides full transparency on this process.

Author's Role (Richard Kent Gates):

- Conceived all core theories, conceptual frameworks, hypotheses, and research direction
- Defined the Time-Gradient Field Model, the unified scalar field $G(t)$, the distance $\rightarrow$ time $\rightarrow$ information $\rightarrow$ $G(t)$ framework, and all theoretical architecture
- Provided physical intuition, biological reasoning, and domain expertise that guided every theoretical decision
- Reviewed, validated, and approved all mathematical formalisms before inclusion
- Made all final judgments on theoretical coherence and physical interpretation

AI Research Partners:

- **Mimo V2.5 (opencode platform)** — Primary mathematical formalization partner. Assisted in translating conceptual physics into mathematical notation, generating numerical proof scripts, compiling LaTeX documents, and building the website infrastructure.
- **Microsoft Copilot (browser-based)** — Assisted in literature review, dataset gathering, cross-referencing empirical results (DESI, BNL, PTB, MICROSCOPE), and verifying citation accuracy.
- **Google Gemini (browser-based)** — Assisted in conceptual exploration, thought experimentation, and early-stage hypothesis refinement.

Nature of AI Involvement:

The AI tools functioned as research assistants — analogous to graduate students or technical collaborators who help formalize, compute, and organize ideas that originate from the principal investigator. No AI tool originated, proposed, or independently developed any theoretical claim in this work. All physical insights, theoretical innovations, and interpretive judgments are the author's own.

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